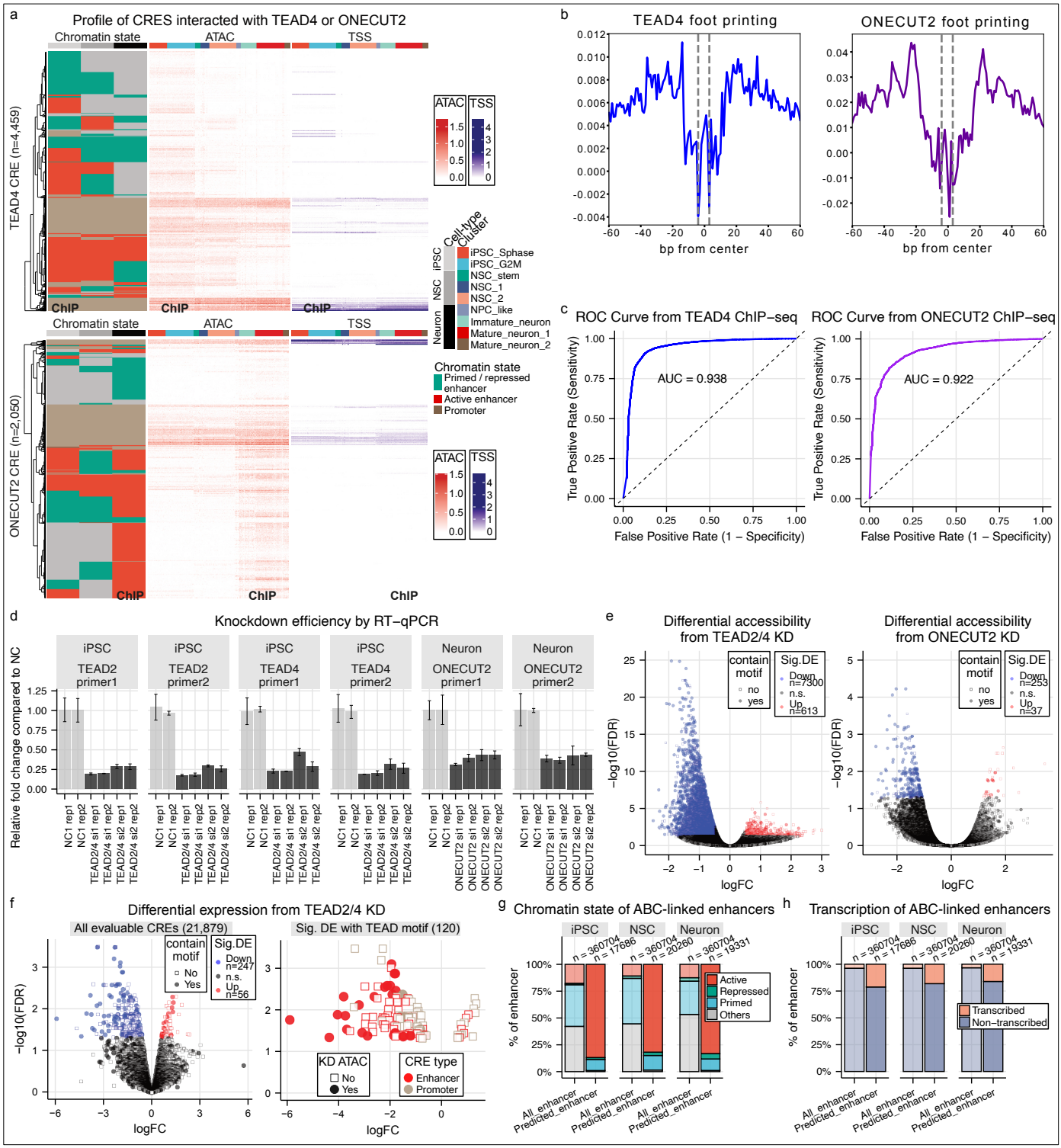


Extended Figure 4 | Validation of TEAD4 and ONECUT2 regulation using ChIP-seq and knockdown experiments.

a, Heatmaps showing the chromatin states across the three samples, and ATAC and TSS signals across metacells for CREs associated with TEAD4 (upper) and ONECUT2 (lower). Associated CREs were defined by significant ChIP-seq binding and presence of corresponding TF motifs (TEAD[1-4] for TEAD4, ONECUT[1-3] for ONECUT2). **b**, TOBIAS-corrected ATAC signals at footprinting-positive CREs. Representative results are shown from TEAD4 in the iPSC_Sphase cluster and ONECUT2 in the Immature_neuron cluster. **c**, Receiver operating characteristic (ROC) curves evaluating TOBIAS footprinting scores using ChIP-seq binding as the reference. Results are shown for TEAD4 in the iPSC_Sphase cluster and ONECUT2 in the Immature_neuron cluster. **d**, RT-qPCR validation of siRNA-mediated knockdown of TEAD2/4 in iPSCs and ONECUT2 in neurons, using two independent primer sets for each gene. Relative expression levels are shown compared with the non-targeting control siRNA (NC1). **e**, MA plots showing differential chromatin accessibility between TF knockdown and non-targeting control conditions across defined CREs with at least 100 counts in each replicate. Black dots indicate CRE without significant differential accessibility, while filled circles and hollow squares indicate differentially accessible CREs with and without the corresponding TF motifs, respectively. Differential accessibility was assessed using edgeR. **f**, MA plots showing differential CRE transcription following TEAD2/4 knockdown in iPSCs, measured by CAGE. Left: All evaluable CREs (n = 21,879), highlighting significantly differentially transcribed CREs containing TEAD motifs. Right: The subset of significant differentially transcribed TEAD motif-containing CREs (n = 120), highlighting their promoter classifications and significant accessibility reduction from TEAD knockdown. **g**, Chromatin state distributions of all enhancers and enhancers predicted to be linked to target genes by the ABC model (ABC score > 0.02) across the three cell types. **h**, Transcriptional state distributions of enhancers stratified as in (**g**).